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Mode and Dispersion of the Length of Life

Väinö KANNISTO*

The changes which characterize the on-going stage of mortality transition – called the era of delayed ageing – are usually measured by such variables as age-specific death rates, life expectancy at birth or at some other age, probabilities of dying and of survival, to mention a few. No convenient summary measure has, however, been accepted. A disadvantage of the measures used is that, in order to be specific to ageing, they require an arbitrary selection of age limits.

Life expectancy at birth which holds center stage in the description of mortality, is for our purpose particularly ill suited, dependent as it primarily is on what happens in infancy and youth. The saving of a young life may add 60-80 years to the numerator while a life saved in old age may add only a few. Even a careful observer of a time series of the life expectancy at birth will have difficulty in discerning the different stages of transition and will most likely not notice at all the shift to the present stage.

In contrast, the modal length of life offers a good basis for the observation of survival to old age and of longevity. It has a relevant conceptual underpinning as it represents the most common, or typical, length of life under a given mortality regime, free from any arbitrary age limit. Though little used in contemporary demography, it has been more highly valued in the past.

Lexis divided the length-of-life distribution ($d(x)$ in a life table) into three parts: (1) a J-curve right after birth which reflected defects in the human constitution; (2) a normal curve around the mode in late life; and (3) the area between, representing premature deaths, ascribed to abnormal conditions or external influences. Where (2) and (3) overlapped, he attributed a fraction of deaths, declining with age, to the abnormal circumstances causing premature death. This left him with (2) as the typical or natural lifetime, expressed by the error law (Lexis, 1877). The 1881-90

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curve in Figure 1 gives an approximation of what Lexis saw while the 1991-95 curve shows the situation as it generally is today in low-mortality populations. The late life mode is now much more pronounced – much more common – and at a little higher age. The life expectancy at birth has moved right, closer to the mode.

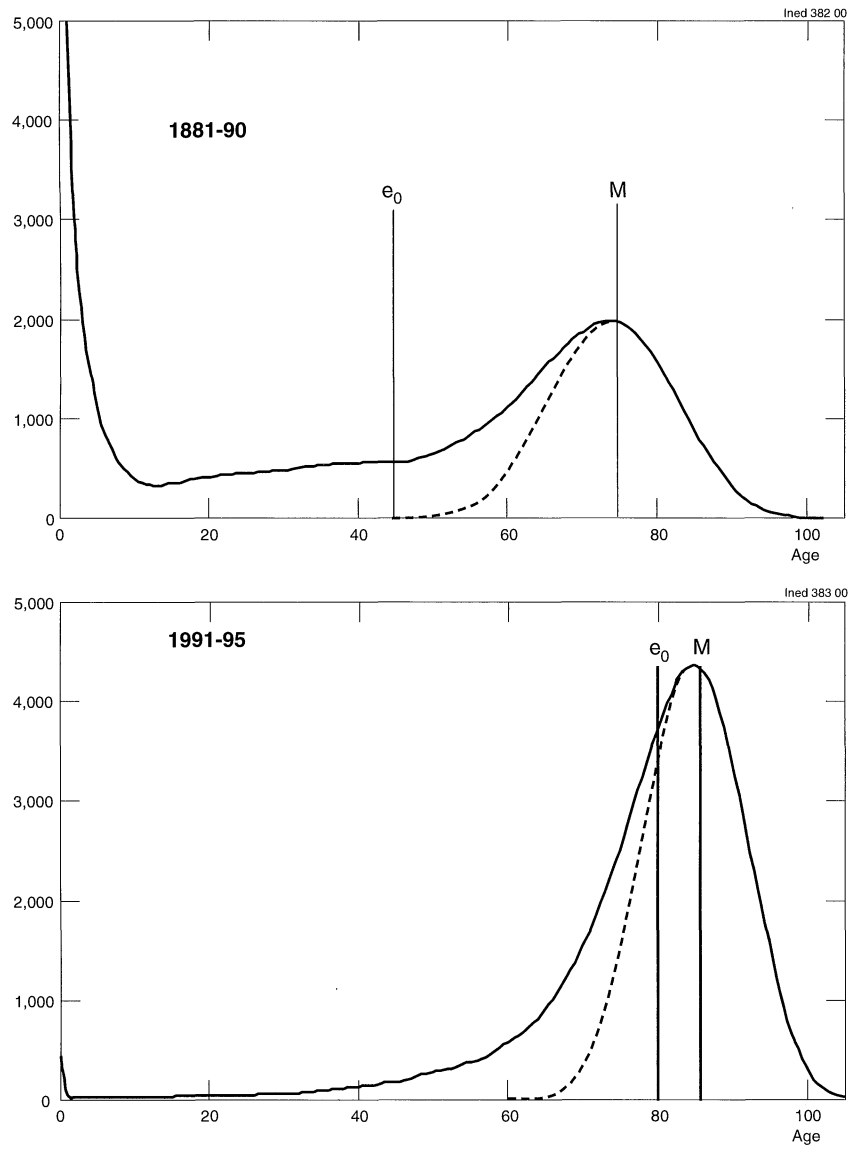


Figure 1.– Length-of-life distribution before and after a century of transition. Finland, female

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Lexis found that above age 60, the actual mortality curves were in reasonable accord with theoretical predictions based on a normal life span ranging from 67 years for Belgian men to 75 years for Norwegian and Swedish women and a probable error of six or seven years. Only in Norway, however, did the normal group comprise more than half the total population (Lexis, 1877, pp. 41-64).

Later, using data for the early 1880s, Lexis calculated the natural lifetime to be between 70 and 72.5 years in Central European countries, 75 in Sweden and 78 in Norway (Lexis, 1903, pp. 112-116).

His example was followed by Ballod who found that the normal lifetime in Germany was 68 years in the cities, 73 in rural areas and 77-78 in the most purely agricultural regions. He thought that the "ideal lifetime" would hardly be much more than 80 years (Ballod, 1899, pp. 57-58). It is worth noting that these writers did not see the normal lifetime in a strictly biological sense but recognized differences caused by environment.

Still in the same tradition, Dublin constructed a "hypothetical life table" according to conditions which were "about to be reached" in the 1920s and found the mode in late life to be between 75 and 80 years (Dublin, 1923, p. 191).

Another approach towards determining the normal or attainable lifetime is to separate the underlying causes of death into preventable and non-preventable. Undoubtedly the best-known and most successful work in this line was the division of causes of death into endogenous and exogenous by Bourgeois-Pichat (1952). His work gave much new insight into the problem but the actual division presented problems because death may result from the interaction of both endogenous and exogenous conditions (Gavrilov and Gavrilova, 1991, pp. 107-110). To add an example to those mentioned by these authors, accidents are usually considered to be among the most purely extraneous causes but accidental falls of old persons are usually due to their age. It is also difficult to extend the method of Bourgeois-Pichat to the highest ages because of lack of data. Following a World Health Organization recommendation from the 1950s, most countries end their cause-of-death cross-classification with age giving an open-ended group "85 and over".

We therefore opt for an all-cause approach to the measurement of the usual lifetime. Significant for our choice of a basic indicator is the historical development which illustrates in Figure 2 the specificity of the mode vs. life expectancy at birth. Before the great secular transition, mortality was high with sharp short-term fluctuations visible even in decennial averages. Both the mean and the mode were affected in the same manner. In the 1880s, the mean life expectancy began to increase in Finland and the fluctuations tapered off. Also the mode became smoother but without an appreciable rise in level. Ever since then, the mean has been increasing monotonically while only a slight rise in the mode could be recorded for women and none at all for men until around 1970 when it suddenly began

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to rise significantly for both sexes. A new era, not visible in the mean, had begun.

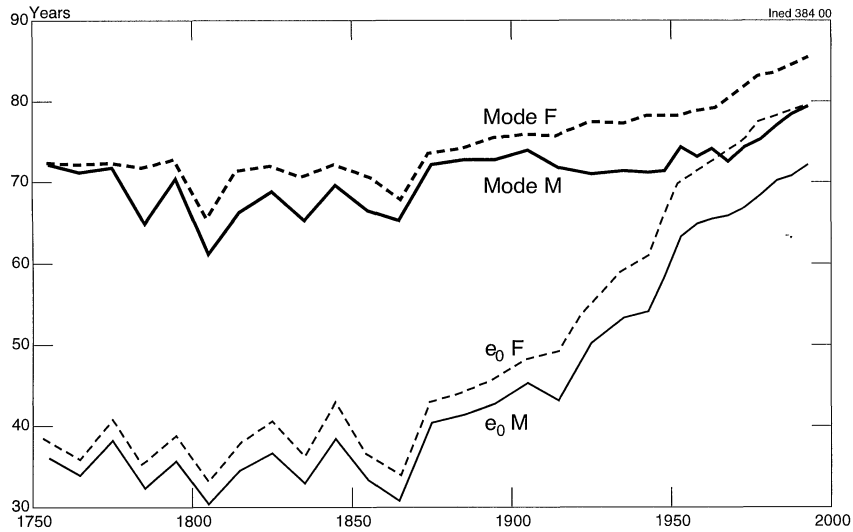


Figure 2.– Modal and mean length of life in Finland, 1750-1995

The late life mode gains additional significance from being remarkably steady over time and space, close to the biblical "three score and ten". Its recent increase in low-mortality countries is entirely unprecedented.

We shall therefore use the mode and its derivatives as indicators of the typical length of life and its dispersion.

The length-of-life distribution is typically bi-modal with one mode in infancy and another in late life while additional modes are sometimes observed at young adult ages. The infancy mode, though drastically reduced, remains universal and its shape has become ever sharper. With modal length of life we shall in the following understand the late life mode which is easy to determine as we know of no case where there has been more than one true mode after middle age. The presence of two or more modes in late life would indicate that the basic data are either insufficient or affected by age heaping or other defect.

Following the concept of Lexis, we would have to assume that premature deaths occur, in diminishing numbers, also at and above the modal age. However, for lack of means to measure them we desist from the attempt.

The mode and related parameters can of course be calculated both in cohort and period life tables. Examination of the recent unprecedented decline in old age mortality in a large number of countries with good quality

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data has shown that it was caused by period factors and not by supposedly healthier cohorts advancing in age (Kannisto, 1994, pp. 55-59). We therefore prefer to examine period life tables in the following.

Due to a certain flatness in the $d(x)$ curve near its highest point, the mode is sensitive to minor variations in the curve which makes it important that the curve is unimodal and relatively smooth. If the basic life table does not satisfy this requirement, it is advisable to smooth it. However, excessive smoothing should be avoided because it reduces the height of the mode. A one-year shift in the mode without appreciable change in the $d(x)$ curve would translate into a change of about half a year in the dispersion parameter, a higher mode being accompanied by reduced dispersion and vice versa.

In order to ensure sufficient precision in the various parameters, the mode needs to be determined in fractions of year. We have done this according to the following formula:

$$\mathrm{M} = x + \frac{d(x) - d(x-1)}{[d(x) - d(x-1)] + [d(x) - d(x+1)]} \qquad [1]$$

where x is the age with the highest d in late life.

We then proceed to calculate the *standard deviation of individual life durations above the mode*, SD(M+) under the assumption that deaths are evenly distributed within each year of age. SD(M+) is the root-mean-square of those deviations from M which are positive.

Table 1 gives this and some other longevity parameters for 15 countries according to the most recent life tables, generally from the 1990-95 period. The ranking of the countries by mode differs to some extent from that by life expectancy at birth. A large difference between the two – as for French males and U.S. males and females – is caused by relatively high mortality in either childhood, youth or middle age. The females have in each country a survival advantage over males both in the mean and the modal length of life, the difference in the former indicator being an average 7.07 years, in the latter 5.76 years.

The dispersion of individual lifetimes, measured by the standard deviation above the mode, varies in this group roughly between 6 and 9 years. Lexis had found, as mentioned earlier, a probable error of 6 to 7 years for the normal life span. The standard deviation is correlated, though not very tightly, with the mode as illustrated in Figure 3, a high modal age being generally accompanied by reduced dispersion. The correlation coefficient r is for males –0.47, for females –0.24.

Next we give in Table 1 another parameter which likewise declines as the mode increases, namely the life expectancy at mode, $e(\mathrm{M})$. The correlation of $e(\mathrm{M})$ with SD(M+), shown in Figure 4, turns out to be extremely rigorous even in a historical perspective. The *lowest* correlation coefficient +0.954 was found for the period 1841-1899 while those for the

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TABLE 1.– LONGEVITY DATA BY COUNTRY, CA. 1990-1995.
IN DESCENDING ORDER OF MODE

	$e(o)$	Mode	SD(M+)	$e(M)$	$\frac{SD(M+)}{e(M)}$	$d(M)$	LFL*	$e(80)$
	Male							
Japan	76.38	83.48	6.96	5.62	1.238	3824	70.4	7.13
France	72.94	82.67	7.25	5.84	1.241	3264	63.5	6.48
Italy	74.06	82.40	7.10	5.73	1.239	3512	65.4	6.72
Switzerland	74.19	82.26	7.16	5.85	1.224	3536	68.7	6.78
USA	72.12	81.24	8.09	6.57	1.231	3130	67.7	7.07
Austria	72.48	81.11	7.22	5.84	1.236	3436	65.4	6.28
Norway	74.00	80.63	7.85	6.41	1.225	3564	73.9	6.53
Sweden	73.55	80.17	7.73	6.24	1.239	3568	74.3	6.31
Netherlands	74.10	79.94	8.01	6.43	1.246	3628	76.8	6.34
Finland	72.13	79.37	8.11	6.54	1.240	3318	72.9	6.30
Belgium	72.43	79.34	7.93	6.38	1.243	3493	73.5	6.13
Denmark	72.62	79.27	8.32	6.73	1.236	3282	73.7	6.44
England	73.41	79.22	8.31	6.74	1.233	3500	77.7	6.44
Slovenia	64.90	78.48	8.04	6.51	1.235	3074	66.2	5.93
Poland	68.12	76.43	8.93	7.35	1.215	2848	70.5	6.00
	Female							
Japan	82.85	88.22	6.60	5.28	1.250	4550	77.7	9.47
France	81.15	87.88	6.16	4.90	1.257	4446	71.7	8.83
Switzerland	81.05	87.63	5.97	4.83	1.236	4472	71.1	8.62
USA	79.05	87.53	6.90	5.60	1.232	3564	65.6	9.17
Norway	80.09	86.76	6.40	5.06	1.265	4138	69.9	8.25
Netherlands	80.20	86.67	6.55	5.18	1.264	4190	72.2	8.38
Italy	80.53	86.61	6.14	5.00	1.228	4484	73.6	8.19
England	78.96	86.17	7.02	5.65	1.242	3718	67.5	8.41
Denmark	77.82	85.57	6.92	5.59	1.238	3584	66.2	8.15
Sweden	79.53	85.42	6.56	5.36	1.224	4404	76.0	7.92
Austria	79.04	85.39	6.25	5.04	1.240	4528	72.3	7.51
Belgium	79.13	85.33	6.80	5.44	1.250	4161	74.4	7.88
Finland	79.71	85.32	6.55	5.25	1.248	4362	75.0	7.72
Slovenia	77.76	84.64	6.03	4.88	1.236	4208	69.8	7.07
Poland	76.57	83.16	6.89	5.60	1.230	4026	74.2	6.96

* Lives of full length, per cent.

later periods were higher still and that for all periods combined r was +0.995. Calculated for the 15 countries in Table 1 it was +0.996 for men and +0.983 for women.

At this point it is of interest to recall that in the normal curve, the ratio of the standard deviation to the mean deviation equals the square root of one-half of π which gives the constant 1.253. In our data above the mode, the deviations are represented respectively by SD(M+) and $e(M)$, and allow us to calculate the corresponding ratios for all countries in the table. These vary only within narrow limits. The average ratio is 1.235 for men and 1.243 for women and is in only rare cases less than 1.23 or more

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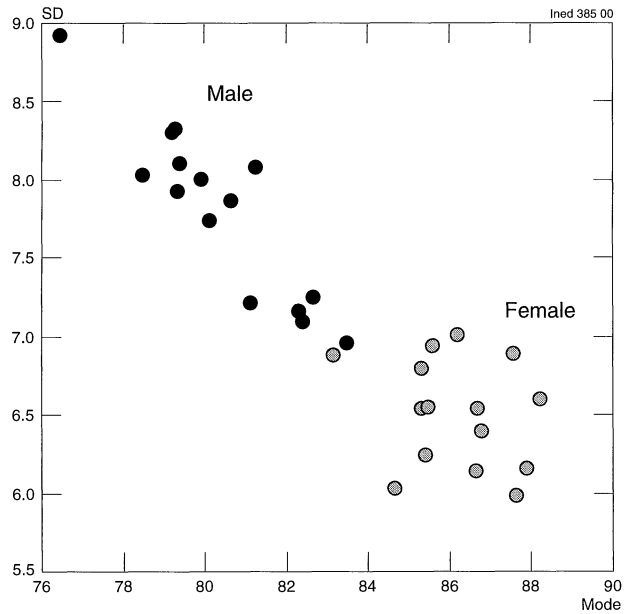


Figure 3.– Mode and standard deviation above mode.
15 countries, most recent period

than 1.25. It is interesting to note that these empirical ratios from a number of countries are very close indeed to the ratio 1.253 of a normal curve. Such steadiness is not intrinsic to this ratio which could conceivably show much greater variation. Rapidly rising mortality would lower both SD(M+) and e(M) but would lower the numerator more and result in a lower ratio. A slower ascent of mortality would increase both indicators but again the numerator relatively more, thus increasing the ratio. Experiments with various plausible mortality trajectories indicate that the differences could be significantly larger than those in Table 1. The fact that such differences have not been encountered in empirical data, testifies to the existence of a fairly universal mortality pattern in old age which is consistent with the observation of Lexis that the distribution of deaths above the mode approximates the normal curve. Our data point to only a slight tendency toward a more compressed form than the side of a normal curve.

The demonstrated steadiness of SD(M+)/e(M) makes it feasible to use e(M) – which is easy to determine in a life table – as proxy for SD(M+) applying a simple multiplier such as 1.25.

The level of concentration of individual life durations around the mode is seen in d(M) which gives the number, out of 100,000 newborn, who die during this single year of age. In males it is in the leading countries between 3,500 and 4,000, in females considerably higher, reaching

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4,500 in some cases. In the 19th century it was almost without exception well below 2,000 but, of course, very high for infants. Another indicator of concentration is the column LFL which shows the percentage of lives which end under the symmetrical curve where the dispersion above the mode is replicated in a mirror image below it. Following the concept of Lexis, these might be considered lives of typical or full length under a given mortality regime. As mentioned earlier, Lexis found that in late last century, only in Norway did this typical group make up more than half the total population. Today this proportion stands in low mortality countries between two-thirds and three-quarters. In the 15 countries in Table 1 it is slightly higher for women than men: 71.8 vs. 70.7 but it should be kept in mind that this typical length is higher for women than men.

There exists some controversy among demographers as to whether mortality, specifically in old age, is being compressed into a shorter age span or not. Myers and Manton (1984) present data on deaths at ages 60 and over in the United States in 1962-1979 and conclude that steady increase in the standard deviation of the age at death above 60 years "provides clear evidence that mortality compression did *not* occur at older ages over this period". Citing some unreliability in the U.S. data, the authors used the "numerator method" in which deaths are not related to any population. The main problem, however, is that when measured above a fixed age limit at a time when life expectancy is growing and the bulk of the *d*-

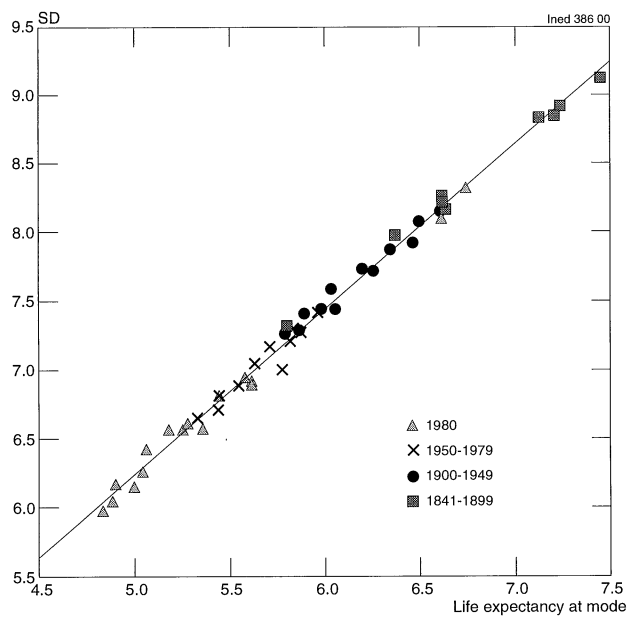


Figure 4.– Life expectancy at mode and standard deviation of length of life above mode. 16 countries at various periods. Female

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distribution is moving to higher ages, the dispersion has an in-built tendency to increase. Using the mode instead of a fixed age limit, we find in Table 2 that also in the U.S. as old age mortality has declined, the dispersion of the length of life has been steadily reduced. Even allowing for some uncertainty in the basic data, the tendency looks convincing.

TABLE 2.– MODE AND DISPERSION OF LENGTH OF LIFE. UNITED STATES, FEMALES

Period	Mode	SD(M+)	e(M)	$\frac{SD(M+)}{e(M)}$
1931-35	76.84	8.70	7.06	1.232
1951-55	80.94	7.97	6.57	1.232
1971-75	84.47	7.40	5.97	1.239
1991-95	87.39	6.98	5.65	1.235

Source: Berkeley Mortality Database. Life tables for the United States published by U.S. Department of Health and Human Services.

Applying Keyfitz's measures of mortality entropy to the good quality data for the Netherlands, Nusselder and Mackenbach (1996) find that some absolute rectangularization of the survival curve has recently been taking place there. This means that mortality has been compressed into a shorter age interval.

Searching for evidence in a longer historical perspective, we show in Figure 5 the development of the late life mode since the mid-19th century in four countries with data of good quality. We compare the mode with life expectancy at mode which may be understood as a proxy for dispersion. The curves are drawn between values in successive life tables. Zig-zag movements show occasional periods of stagnation but the general trend is unmistakable and convincing in all cases: as the mode shifts to higher age, the dispersion above it declines. Even the levels of both variables are remarkably similar in the four countries.

On a broader front, further evidence on recent developments in old age mortality is available in the Kannisto-Thatcher Oldest-Old Database at the University of Odense, Denmark, in which mortality data above age 80 have been assembled from 31 countries. Arranged into cohort mortality histories to which the extinct cohort method of Vincent (1951) was applied, the data were assessed as being of good quality in 18 countries, for 13 of which the series reach back to 1950 or earlier.

If we simply calculate in this material the standard deviation of the age at death above the fixed limit of 80 years, we certainly find that it has grown, namely from 6.37 years in 1960-65 to 8.25 years in 1990-95, but this has nothing to do with compression of mortality or the absence of it. The relevant test is to measure dispersion above the mode.

When the data begin at age 80, the mode can only be determined when it equals or exceeds 81 years. This limit was passed by females in the pooled group of 13 countries in 1960, by males in 1990. We can there-

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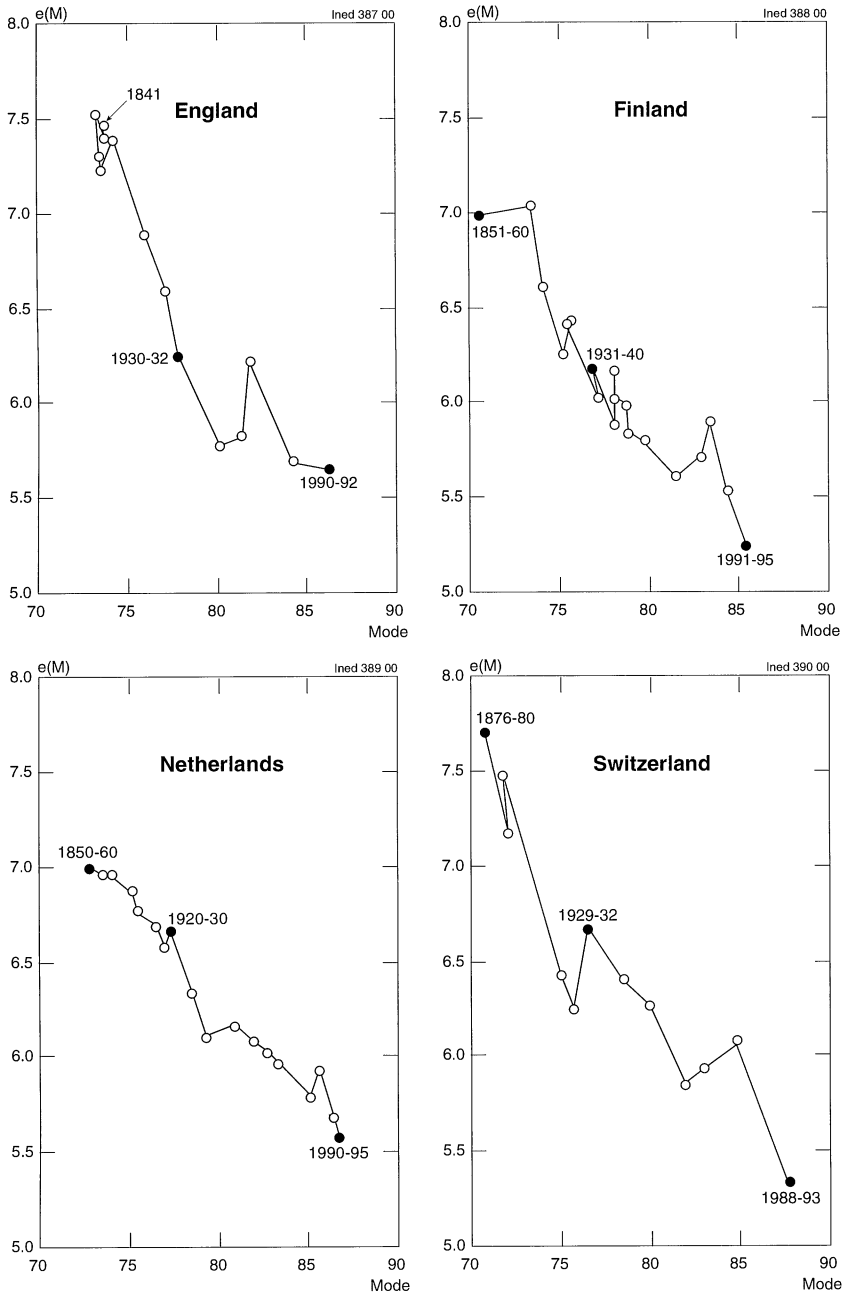


Figure 5.– Mode and life expectancy at mode.
Successive life tables in 4 countries. Female

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fore present results for females in the period 1960-1995 in Table 3. Based on accurate and massive data (215 million person-years of observation above age 80 resulting in 25 million deaths) they show that the steady decline in old-age mortality, as evidenced by an increase in life expectancy at 80, has been accompanied by an equally steady advance of the mode but that at the same time the dispersion of the length of life as well as life expectancy at mode have been declining. This may be interpreted to mean that the prolongation of human life is meeting increasing resistance.

TABLE 3.– MODE AND DISPERSION OF LENGTH OF LIFE.
THIRTEEN COUNTRIES WITH GOOD DATA, FEMALES*

Period**	$e(80)$	Mode	SD(M+)	$e(M)$	$\frac{SD(M+)}{e(M)}$
1960-65	6.28	81.50	7.01	5.68	1.234
1965-70	6.49	82.21	6.93	5.59	1.239
1970-75	6.76	83.09	6.81	5.48	1.242
1975-80	7.12	84.05	6.69	5.38	1.243
1980-85	7.55	84.77	6.71	5.42	1.238
1985-90	8.08	85.83	6.64	5.36	1.238
1990-95	8.56	86.83	6.53	5.27	1.239

* Austria, Denmark, England and Wales, Finland, France, Germany (West), Iceland, Italy, Japan, the Netherlands, Norway, Sweden, Switzerland.
** The rates have been calculated by the extinct-cohort method and are therefore for overlapping periods.

Conclusions

The mode and its upward standard deviation are useful indicators of the length of life and of the dispersion of individual life duration. Unaffected by premature deaths and free from bias caused by a fixed age limit, they describe the typical life span and the life above it under a given mortality regime.

While the mode is increasing, it is not simply sliding to the right. Instead, its right-hand slope is being flattened vertically as if it was meeting an invisible wall. We contend that the ascending trajectory of mortality at these ages indeed forms such a barrier but only in a relative sense, offering stiffer resistance to further progress without setting any definite limit to it. To quote Keyfitz (1978), there is "an uphill road ahead" for extending human life.

This view is compatible with comprehensive studies of low-mortality populations which show that while some decline in mortality has been observed at ages as high as 105-109, the pace of decline has been much faster at relatively younger ages: for women three times as fast at age 80 than at 100 (Kannisto, 1996, p. 112).

Though mortality has thus been compressed into shorter age intervals, no realistic scenario would assume that it ever would be reduced into

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a single point in age. All biological properties display some degree of individual variation and, in human longevity, this heterogeneity is definitely a factor (Vaupel *et al.*, 1979; Wilmoth and Horiuchi, 1998). In our opinion, heterogeneity should not be seen as a mainly congenital attribute, genetic or otherwise. A genetic element is certainly present though it has never been well quantified. On the other hand, it has been shown that frailty is not permanent through a person's lifetime but varies according to environment (Kannisto, 1992). Human capacity for recuperation should not be underestimated.

While there is still lack of proof of substantial congenital heterogeneity, evidence of profound social heterogeneity has been produced in many countries. This problem complex has been very thoroughly researched in Finland where the entire population is classified by socio-economic status, educational level and other variables since 1970 and the path of each individual until death is followed through extensive record linkage. The thrust of these studies is that while mortality has been declining in each social group, differences between them have not narrowed down (Valkonen *et al.*, 1993; Martelin, 1998). If and when efforts to reduce inequality in face of death will have success, the dispersion of the lifetimes will be further reduced.

Continued observation of the mode and of the dispersion above it over time and space may give clues as to any limits to human life and to the length-of-life distribution that may be expected. Of particular importance is to observe whether the on-going unmistakable compression of mortality continues.

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